

Oscar Jaroker

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Education

Colgate University, BA in Applied Mathematics and Physics Aug 2023–May 2027

- GPA: 3.96 / 4.00 Overall; 4.00 / 4.00 in Majors
- **Coursework:** Data Analysis, Statistical Learning, Combinatorics, Real Analysis, Lagrangian Mechanics, Nonlinear Dynamics & Chaos, Quantum Mechanics, Electronics, Object-Oriented Programming
- **Awards and Honors:** Edwin Foster Kingsbury Prize for “most outstanding” physics student (May 2024)
Osborn Mathematics Prize for achieving the highest score on a 3rd-year level mathematics exam (May 2026)

Experience

High-Performance Computing Intern, National Security Agency, Catonsville, MD Summer 2026

- Co-developed a multithreaded C file-transfer tool slated to replace the incumbent HPC transfer utility, outperforming every alternative benchmarked; raised sustained throughput over a 200 Gbit/s InfiniBand fabric by 29% over the incumbent on RAM-backed storage and 32% on a Lustre parallel filesystem
- Designed the tool’s shipped congestion-control algorithm, keyed on delivery rate and minimum RTT rather than packet loss: 260% higher throughput than TCP CUBIC on a lossy emulated WAN and 74% lower transfer time on high-bandwidth-delay paths, eliminating a throughput-collapse failure mode
- Delivered weekly progress presentations to interns and staff; co-authored the project’s technical paper and a second paper documenting the team’s AI-assisted software engineering workflow, for use as an internal resource

Academic Research and Teaching Roles, Colgate University, Hamilton, NY

- **Undergraduate Research Assistant**, Geography Laboratory Sept 2025–May 2026
Built and programmed Arduino-based data collection devices for actuator movement; integrated CH₄ and CO₂ sensors with I²C communication protocols; design and code at github.com/ojaroker/AQUA_Flux_Colgate
- **Teaching Assistant**, Digital and Analog Electronics; Intro Quantum Mechanics & Special Relativity Jan 2026–May 2026
Supported laboratory experiments and circuit troubleshooting; led homework discussion sections and held office hours

Corporate Treasury Intern, Pfizer, New York, NY Summer 2024

- Coded Python API interface to Bloomberg financial data to programmatically generate financial research of the pharmaceutical industry, including drug-industry benchmarking and peer-group metrics dashboards
- Developed pricing simulator for interest-rate swaps using R Programming Language, modeling interest rate movements versus Pfizer’s portfolio

Coding, Math and Physics

- Art of Problem Solving, Millburn, NJ. Summer 2025 - Science Olympiad assistant in optics, circuits, mechanics, robotics, electricity and magnetism. Led 18-student team in problem-solving, collaboration and critical thinking
- International Ivy, Summit, NJ. Summer 2023 - Arduino hardware and software assistant for summer technical camp. Awarded bonus as a top instructor.

Academic Projects & Activities

Koopman Analysis of Chua’s Circuit – oscar.jaroker.com/projects/koopman/ April 2026–May 2026

- Implemented and collected data from Chua’s Circuit, then performed analysis by estimating the Koopman operator with Extended Dynamic Mode Decomposition

President of Colgate Engineering and Design Club February 2025–Present

- Founded and led 40+ member engineering club; secured \$3,000 in project funding for the Spring 2026 semester

RIT Physics Olympiad 2024 & 2025

- Won second place out of 23 teams in 2025 and received a cash prize

Skills

Programming: Python (Pandas, Matplotlib), Java, MATLAB, R (tidyr, ggplot2), SQL, C, C++ , Mathematica

Language: Spanish (Limited Proficiency)